

Probeklausur Mathe



1

1.1

$$a) a^{-1} = \underline{\underline{\frac{1}{a}}}$$

$$c) \frac{1}{b^{-2}} = \underline{\underline{b^2}}$$

$$d) a^{-1} = \frac{1}{a} + \frac{1}{a} = \underline{\underline{\frac{2}{a}}}$$

$$e) \frac{b^{-1}}{\frac{1}{b}} = \frac{\frac{1}{b}}{\frac{1}{b}} = \underline{\underline{1}}$$

$$\begin{aligned} b) \frac{a^{-1}}{b^{-2} + c^5} &= \frac{1}{a} \cdot \frac{1}{\frac{1}{b^2} + \frac{1}{c^5}} \\ &= \frac{1}{a} \cdot \frac{1}{\frac{b^2 + c^5}{b^2 c^5}} = \underline{\underline{\frac{1}{a} \cdot \frac{b^2 c^5}{b^2 + c^5}}} \end{aligned}$$

2

$$\int_0^a x e^{x^2} dx = \frac{e-1}{2}$$

$$\int_0^a x e^{x^2} dx$$

$$\rightarrow \text{let } u = x^2$$

$$\text{so } \frac{du}{dx} = x^2$$

$$= \int_{u(0)}^{u(a)} x e^u \cdot \frac{1}{2x} du$$

$$\leftarrow \Rightarrow \frac{du}{2x} = dx$$

$$= \int_0^{a^2} \frac{1}{2} e^u du$$

$$= \frac{1}{2} \int_0^{a^2} e^u du = \frac{1}{2} [e^u]_0^{a^2} = \underline{\underline{\frac{1}{2}(e^{a^2} - 1)}}$$

$$\text{so if: } \frac{e-1}{2} = \frac{e^{a^2}-1}{2}$$

$$\text{then: } e-1 = e^{a^2}-1 \Leftrightarrow e^1 = e^{a^2}$$

$$a^2 = 1$$

$$\Leftrightarrow$$

$$a_1 = 1$$

$$a_2 = -1$$

③ a) $z = \frac{1+i}{1-i} \quad | \cdot \frac{1+i}{1+i}$

$$z = \frac{(1+i)^2}{(1+i)(1-i)}$$

$$z = \frac{1+2i-1}{1^2-i^2} = \frac{2i}{2} = i$$

$\leadsto \underline{z = 0 + 1i, \operatorname{Re}(z) = 0, \operatorname{Im}(z) = 1}$

b) $z = \frac{i^0}{0!} + \frac{i^1}{1!} + \frac{i^2}{2!}$

$$z = 1 + i - \frac{1}{2}$$

$\underline{z = \frac{1}{2} + i, \operatorname{Re}(z) = \frac{1}{2}, \operatorname{Im}(z) = 1}$

c) $z = \frac{2+i}{2-i} \quad | \cdot \frac{2+i}{2+i}$

$$z = \frac{(2+i)^2}{(2+i)(2-i)} = \frac{4+4i-1}{4-i^2} = \frac{3+4i}{5}$$

$\underline{z = \frac{3}{5} + \frac{4}{5}i, \operatorname{Re}(z) = \frac{3}{5}, \operatorname{Im}(z) = \frac{4}{5}}$

d) $z = e^{-i\pi}$

$$z = \cos(-\pi) + i \sin(-\pi)$$

$\underline{z = -1 + 0i, \operatorname{Re}(z) = -1, \operatorname{Im}(z) = 0}$

e) $\underline{z = 0, \operatorname{Re}(z) = 0, \operatorname{Im}(z) = 0}$

f) $z = \frac{e^{2\pi i}}{(i-1)^{-1}} = \frac{1}{(1-i)^{-1}}$

$\underline{z = 1-i, \operatorname{Re}(z) = 1, \operatorname{Im}(z) = -1}$

④ suppose $f(x) = 4(x-2)^2 + x - 16$

$$= 4(x^2 - 4x + 4) + x - 16$$

$$= 4x^2 - 16x + 16 + x - 16$$

so $f(x) = 4x^2 - 15x$

so $f'(x) = 8x - 15$ and $f''(x) = 8$ and $f'''(x) = 0$

$x_0:$ $0 = 4x^2 - 15x$

$0 = x(4x - 15)$

so $\swarrow \searrow$

$x_{0_1} = 0$ $0 = 4x_{0_2} - 15$

$4x_{0_2} = 15$

$x_{0_2} = \frac{15}{4}$

$P_{ex}: f'(x) = 0 \Leftrightarrow 0 = 8x - 15$

$8x = 15$

$x_{ex} = \frac{15}{8}$

let $P_{ex}(x | f(x)) \Leftrightarrow P_{ex}(\frac{15}{8} | f(\frac{15}{8}))$

$\Leftrightarrow P_{ex}(\frac{15}{8} | -\frac{225}{16})$

$f(x) = 4x^2 - 15x$

$f(\frac{15}{8}) = 4(\frac{15}{8})^2 - 15 \cdot \frac{15}{8}$

$= \frac{1}{4} \cdot \frac{225}{64} - \frac{225}{8}$

$= \frac{225}{16} - \frac{225}{8} = \frac{225 - 450}{16} = -\frac{225}{16}$

$P_i: f''(x) = 0 \Leftrightarrow 0 = 8 \quad \downarrow$

$\rightarrow$ no point of inflection

5

Berechnen der Schnittachse von E_1 u. E_2

$$E_1 = E_2 \Leftrightarrow \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix} + d \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$$

also:

$$\left. \begin{array}{lcl} \text{I} & a + 2b & = -2d \\ \text{II} & 2 + 2b & = -2 - 2c + 2d \\ \text{III} & -b & = -c - d \end{array} \right\}$$

III': $d = -c - b$

↳ in I: $a + 2b = -2(-c - b)$

$$a + 2b = 2c + 2b$$

$a = c$

↳ in II: $2 + 2b = -2 - 2a + 2(-a - b)$

$$2 + 2b = -2 - 2a - 2a - 2b$$

$$2 + 4b = -2 - 4a \Leftrightarrow 4 + 4b = -4a$$

$a = -1 - b$

↳ in E_1 : $\vec{E}_1(a = -1 - b) = \vec{g}$

$$\vec{g} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + (-1 - b) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} -1 + 2 \\ 0 + 2 \\ 0 + 1 \end{pmatrix}$$

$\vec{g} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

Durchstoßpunkt der Schnittachse $\vec{g}$ des E_3 finden:

$$E_3 = \vec{g}$$

$$\begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} + c \begin{pmatrix} 0 \\ -2 \\ -1 \end{pmatrix} + f \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$\left. \begin{array}{l} \text{I} \quad -1 + b = 2 + 2f \\ \text{II} \quad 2 + 2b = 2 - 2c + 2f \\ \text{III} \quad b = -c - f \end{array} \right\}$$

$$\text{III in I: } -1 + (-c - f) = 2 + 2f \\ -1 - c - f = 2 + 2f \Leftrightarrow \underline{c = -3 - 3f}$$

$$\begin{aligned} \hookrightarrow \text{in II: } 2 + 2(-c - f) &= 2 - 2c + 2f \\ 2 - 2c - 2f &= 2 - 2c + 2f \\ 2 - 2f &= 2 + 2f \\ 4f &= 0 \\ f &= 0 \quad \leadsto \underline{c = -3} \quad \leadsto \underline{b = 3} \end{aligned}$$

$$\vec{g}(b=3) = \vec{OS} \\ \vec{OS} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \Leftrightarrow \vec{OS} = \begin{pmatrix} -1+3 \\ 2+6 \\ 0+3 \end{pmatrix}$$

$$\vec{OS} = \begin{pmatrix} 2 \\ 8 \\ 3 \end{pmatrix}$$

$$\Rightarrow \underline{\underline{S(2|8|3)}}$$

6

$$e^{i\varphi} = \cos(\varphi) + i \sin(\varphi)$$

Es sei $f(\varphi) = e^{i\varphi}$ also $f'(\varphi) = i \cdot e^{i\varphi}$ und $f''(\varphi) = -e^{i\varphi}$

und $f'''(\varphi) = -i \cdot e^{i\varphi}$ und $f^{(4)}(\varphi) = e^{i\varphi}$.

Mit $f(\varphi) \approx \sum_{k=0}^4 \frac{f^{(k)}(\varphi_0)}{k!} (\varphi - \varphi_0)^k$ und $\varphi_0 = 0$

folgt $f(\varphi) \approx e^{i \cdot 0} + i e^{i \cdot 0} \varphi - \frac{e^{i \cdot 0}}{2} \varphi^2 - i \frac{e^{i \cdot 0}}{6} \varphi^3 + \frac{e^{i \cdot 0}}{24} \varphi^4$

$$\approx 1 + i\varphi - \frac{1}{2} \varphi^2 - \frac{i}{6} \varphi^3 + \frac{1}{24} \varphi^4$$

$$\approx \underbrace{1 - \frac{1}{2} \varphi^2 + \frac{1}{24} \varphi^4}_{=T(\cos(\varphi)) ?} + \underbrace{i\varphi - \frac{i}{6} \varphi^3}_{\cong T(i \sin(\varphi)) ?}$$



Sei $g(\varphi) = \cos \varphi$

$g'(\varphi) = -\sin \varphi$ $g''(\varphi) = -\cos \varphi$

$g'''(\varphi) = \sin \varphi$ $g^{(4)}(\varphi) = \cos \varphi$

$$T(g(\varphi)) = \sum_{k=0}^4 \frac{g^{(k)}(\varphi_0)}{k!} (\varphi - \varphi_0)^k$$

$T(g(\varphi)) = 1 - \frac{1}{2} \varphi^2 + \frac{1}{24} \varphi^4 = T(\cos(\varphi))$



Sei $h(\varphi) = \sin(\varphi) \Leftrightarrow i \cdot h(\varphi) = i \sin \varphi$

$h'(\varphi) = \cos(\varphi)$ $h''(\varphi) = -\sin \varphi$

$h'''(\varphi) = -\cos(\varphi)$

$T(h(\varphi)) = \varphi - \frac{1}{6} \varphi^3$

also: $T(i h(\varphi)) = i\varphi - \frac{i}{6} \varphi^3 = T(i \sin \varphi)$

Falls $e^{i\varphi} = \cos \varphi + i \sin \varphi$,

dann $e^{i\varphi} \approx T(\cos \varphi) + T(i \sin \varphi)$, mit $T(i h(\varphi)) = T(i \sin \varphi)$

und $T(g(\varphi)) = T(\cos \varphi)$

folgt, dass

$e^{i\varphi} = \cos \varphi + i \sin \varphi$ □

$$\textcircled{2} \quad f(x) = a(x-p)^2 + b(x-p) + c$$

$$\text{Scheitelpunkt } S(2 \mid -1)$$

$$\text{also } f(x) = a(x-2)^2 + b(x-2) - 1$$

$$\text{daher } \underline{p=2}, \underline{c=-1}$$

$$\text{weiterhin } P_{f(1)}(1 \mid 0) \quad \text{und} \quad P_{f(3)}(3 \mid 0)$$

$$\text{also I } 0 = a \overbrace{(1-2)^2}^1 + b \overbrace{(1-2)}^{-1} - 1$$

$$\text{II } 0 = a(3-2)^2 + b(3-2) - 1$$

$$\text{I } 0 = a - b - 1 \quad \Leftrightarrow \quad b = a - 1$$

$$\text{II } 0 = a + b - 1 \quad \text{Ein II: } 0 = a + a - 1 - 1$$

$$2a = 2$$

$$\underline{a=1} \quad \text{und} \quad \underline{b=0}$$